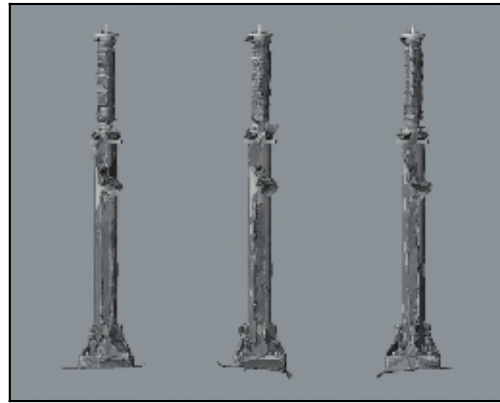


# Why the reconstruction loop broadens: three structural transitions

Each row uses preserved B01–B36 artifacts to show how a new dependency changes the engineering decision, not just the fitted parameters.

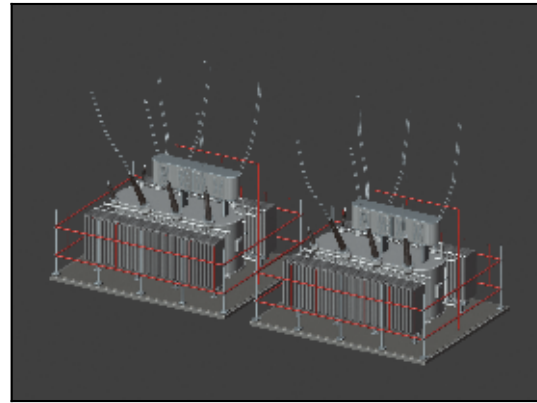
## B01 • local fit



One object: define a local comparison domain, fit pose/scale, then validate.

repetition introduces invariance

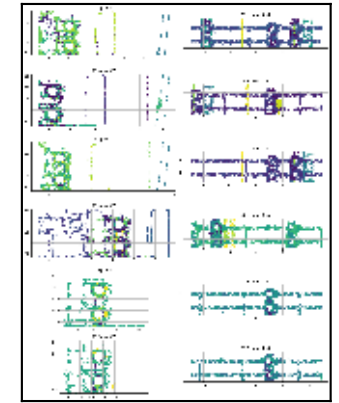
## B08 • shared equipment



Two installations share one transformer MASTER; site routes remain outside the shared asset.

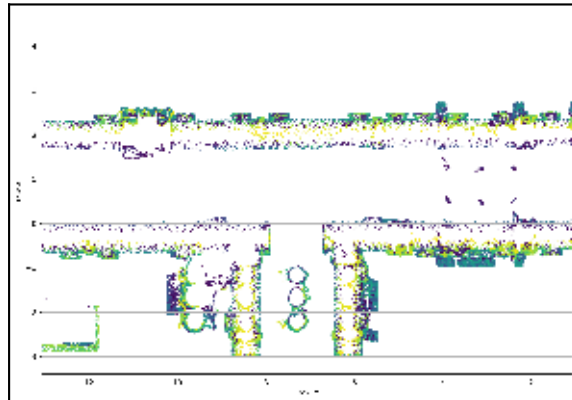
reuse boundary becomes falsifiable

## B15 • revise the reuse boundary



Finite site geometry differs across installations; common spool extent is replaced by site-specific segments.

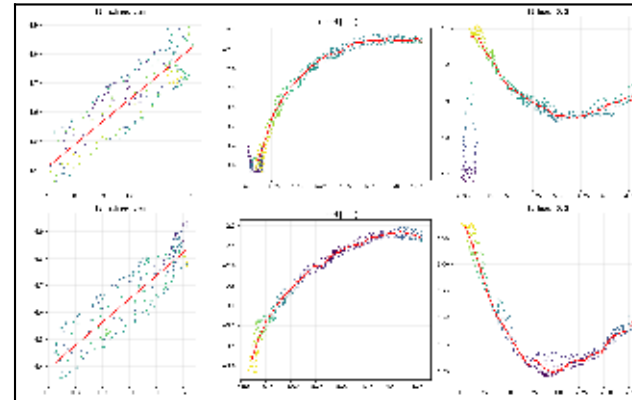
## B20 • ports and routes



Existing equipment ends become explicit ports; one remaining 4100 connection requires a turn.

relationship constraints appear

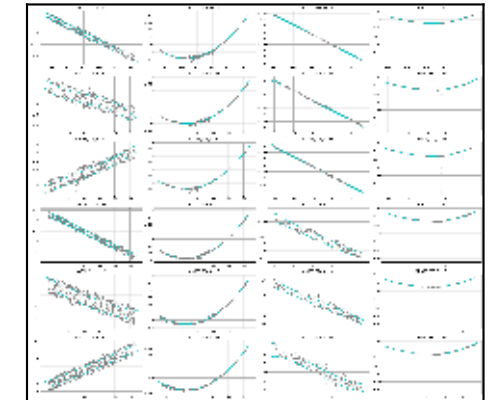
## B23 • change representation class



Profile evidence exposes a mid-span bow; the connection is reformulated from straight to a constrained curve.

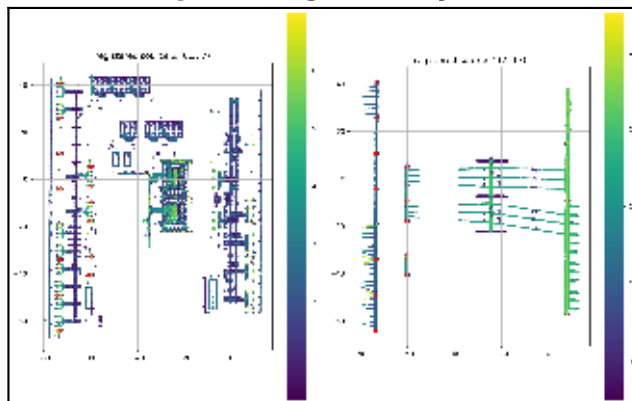
endpoint/topology constraints scale

## B29 • relational constraints at scale



Crossyard conductors are fit while preserving inherited endpoints; continuity and surface fit are checked separately.

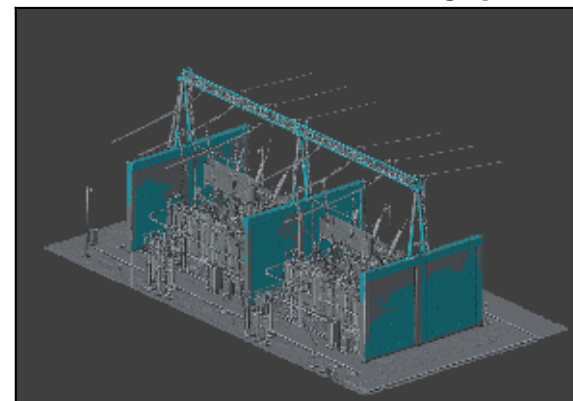
## B25 • unexplained geometry



Registered high structures remain far from the current scene, exposing omissions not captured by inventory status alone.

coverage becomes a task signal

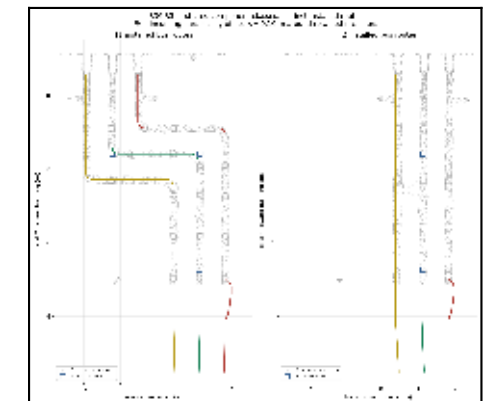
## B26 • reconstruct selected gaps



Coverage-driven follow-up adds missing gantry/firewall structure while retaining the existing station state.

new geometry must fit the existing world

## B36 • state-constrained insertion



Late bus-rack routes must preserve inherited endpoints and reconcile with occupied space and existing infrastructure.

Historical artifacts are shown as evidence of the documented workflow. The sequence is descriptive; it does not establish a universal complexity law or model learning over time.

A Local fit → reusable scope

B Object geometry → connected representation

C Scheduled addition → gap/state-driven closure